

WINNER ENGINEERING LEARNING HUB SERIES

FUNDAMENTAL TOPIC

#03 Pipe Connection



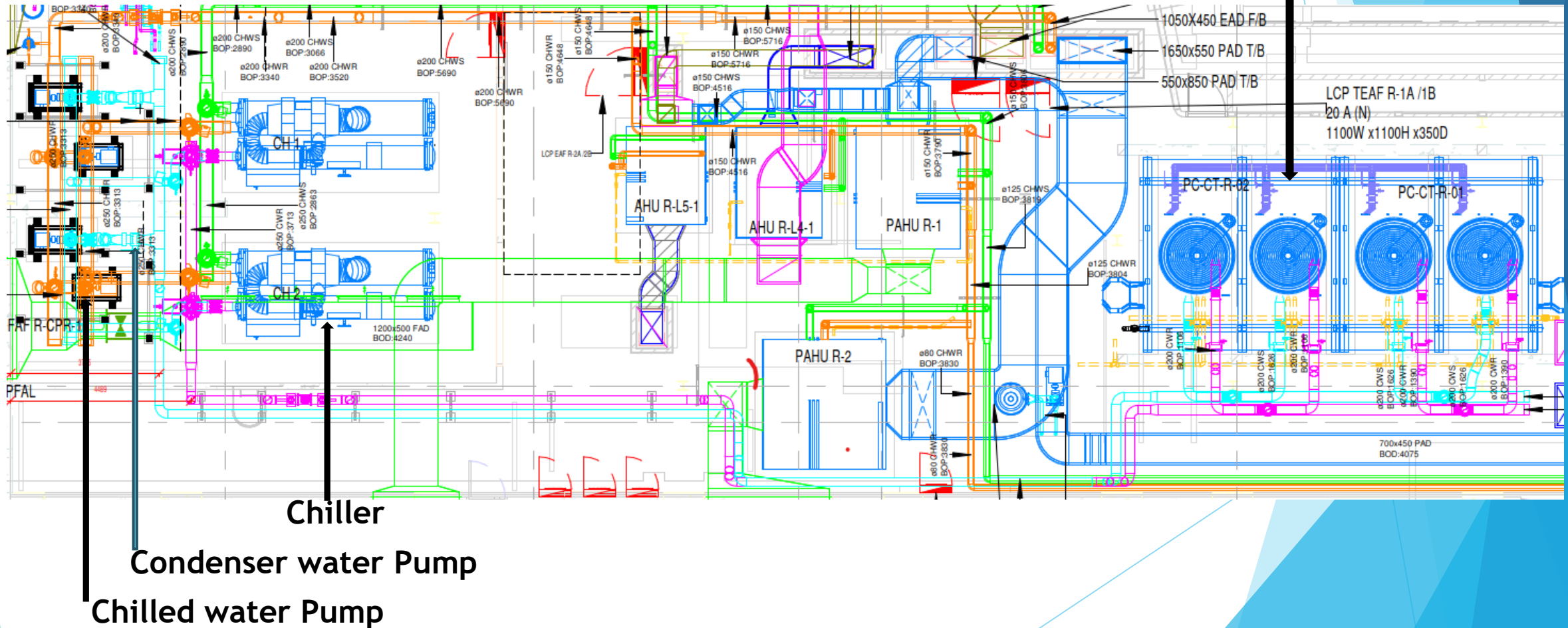
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永胜工程私人有限公司
Excellence in Service

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Chiller Plant Room

- ▶ Chiller plant room is a dedicated space within a building where the equipment and systems for cooling and air conditioning are housed.
- ▶ It is typically located in the basement or Rooftop.



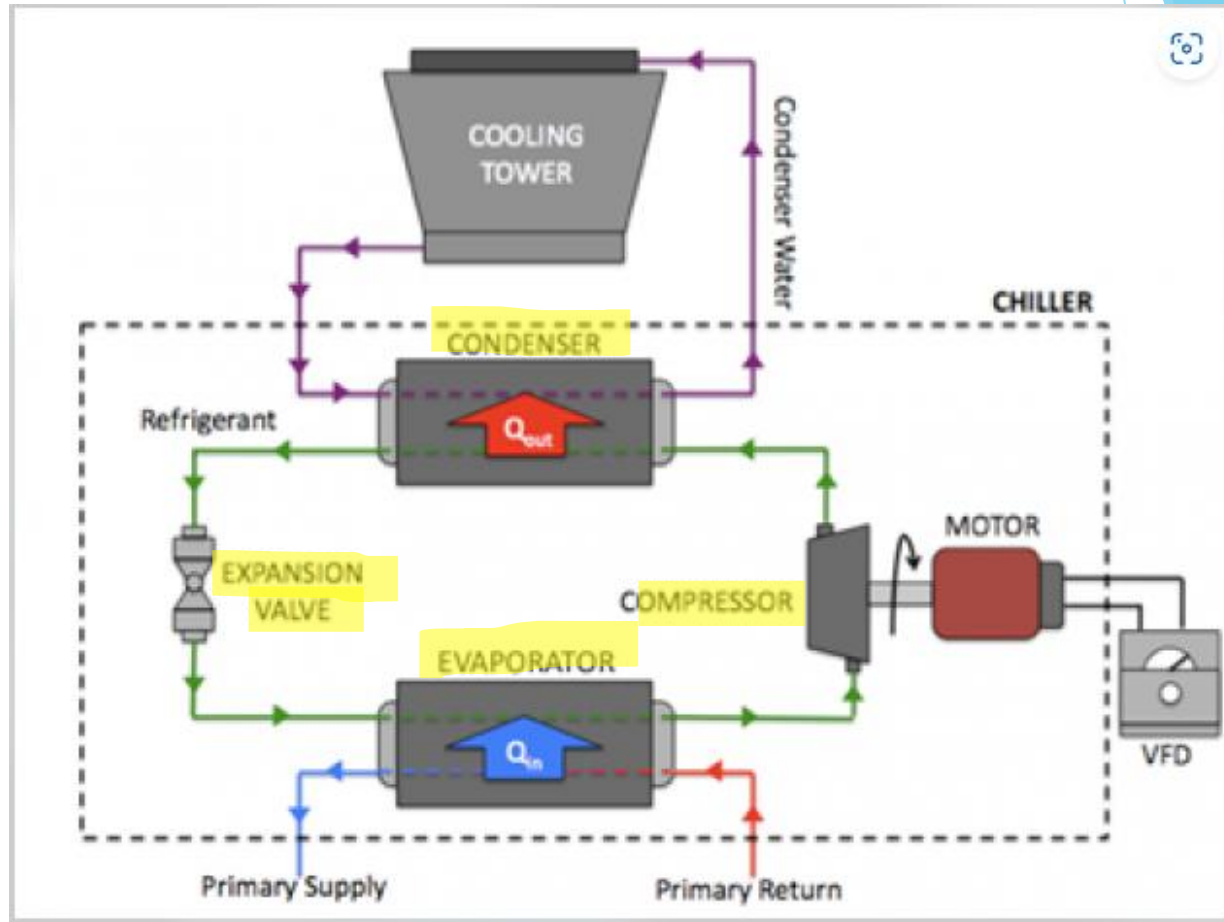
1. Chiller

There are 4 main parts in the Chiller:

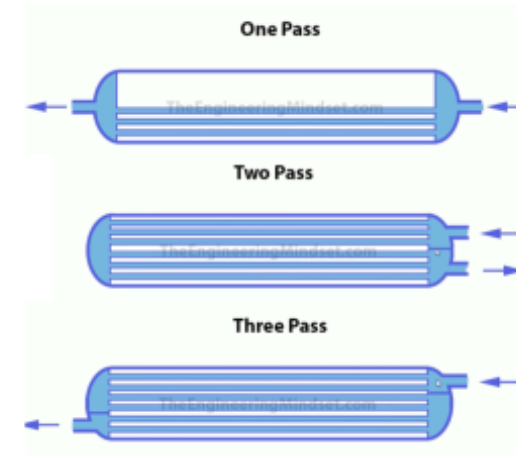
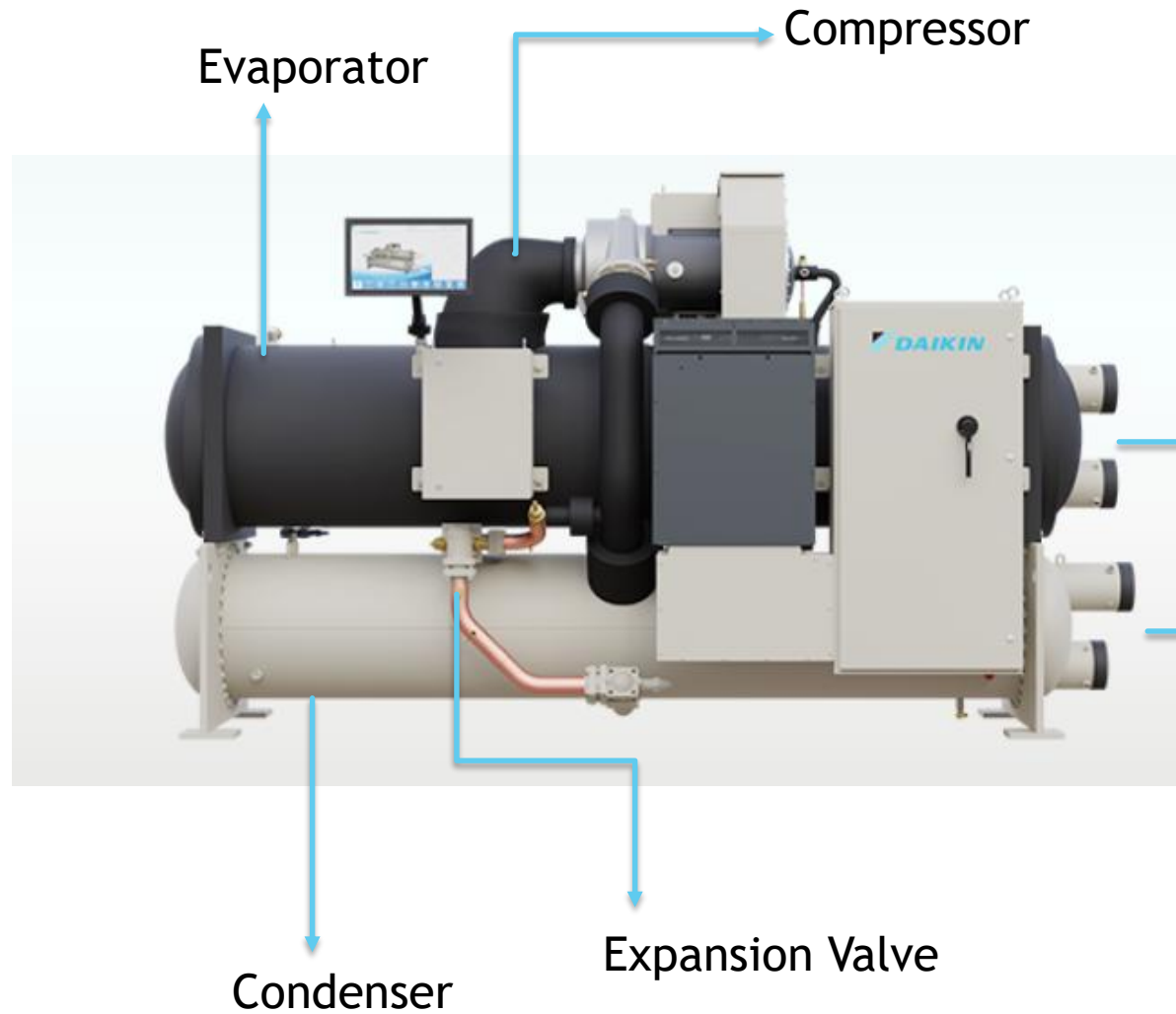
1. Evaporator
2. Condenser
3. Expansion Valve
4. Compressor

And connected with 3 different Loop:

1. Chilled water loop
2. Condenser Water Loop
3. Refrigerant Loop



Main Component of the Chiller



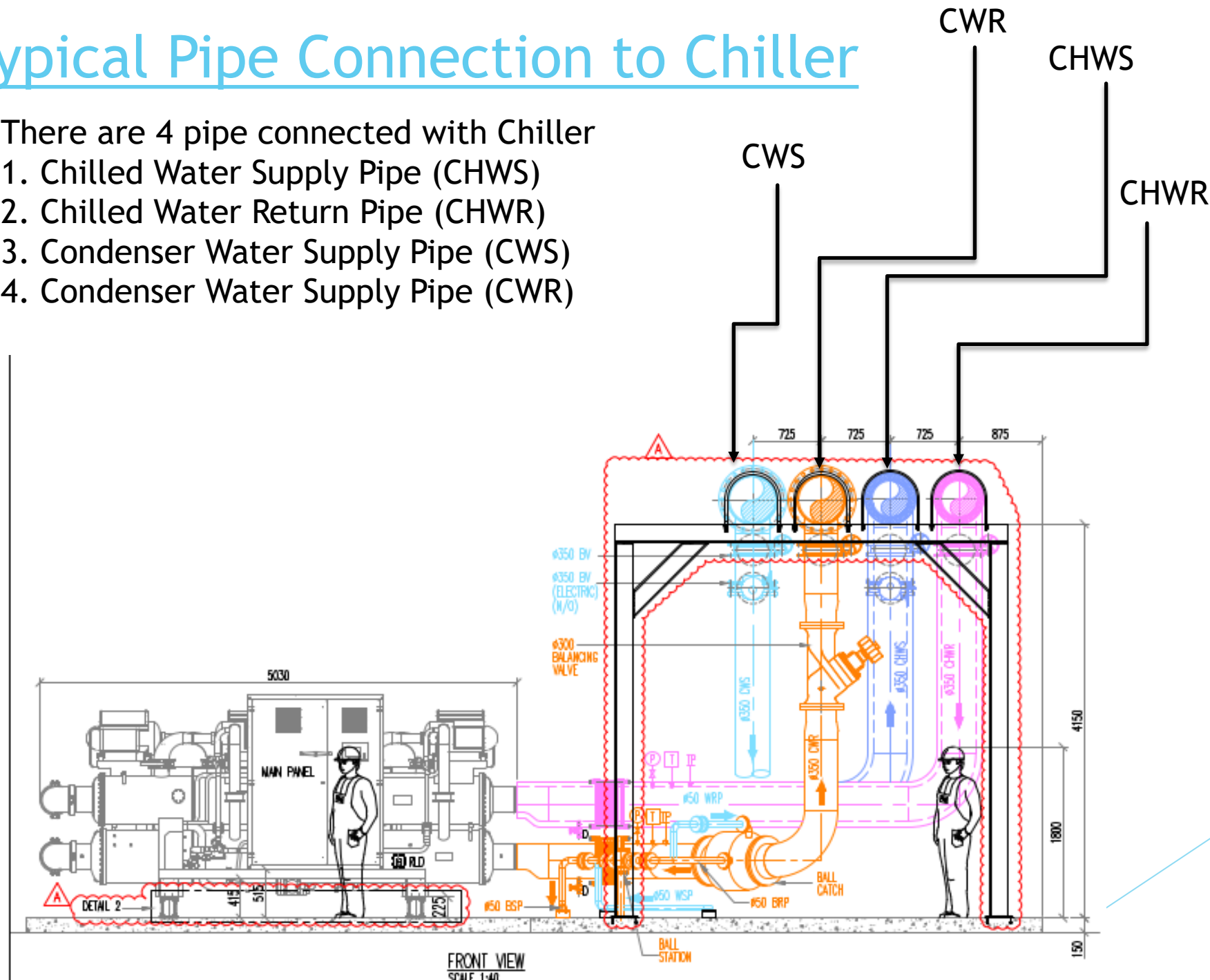
Chilled water inlet & outlet

Condenser Water inlet & outlet

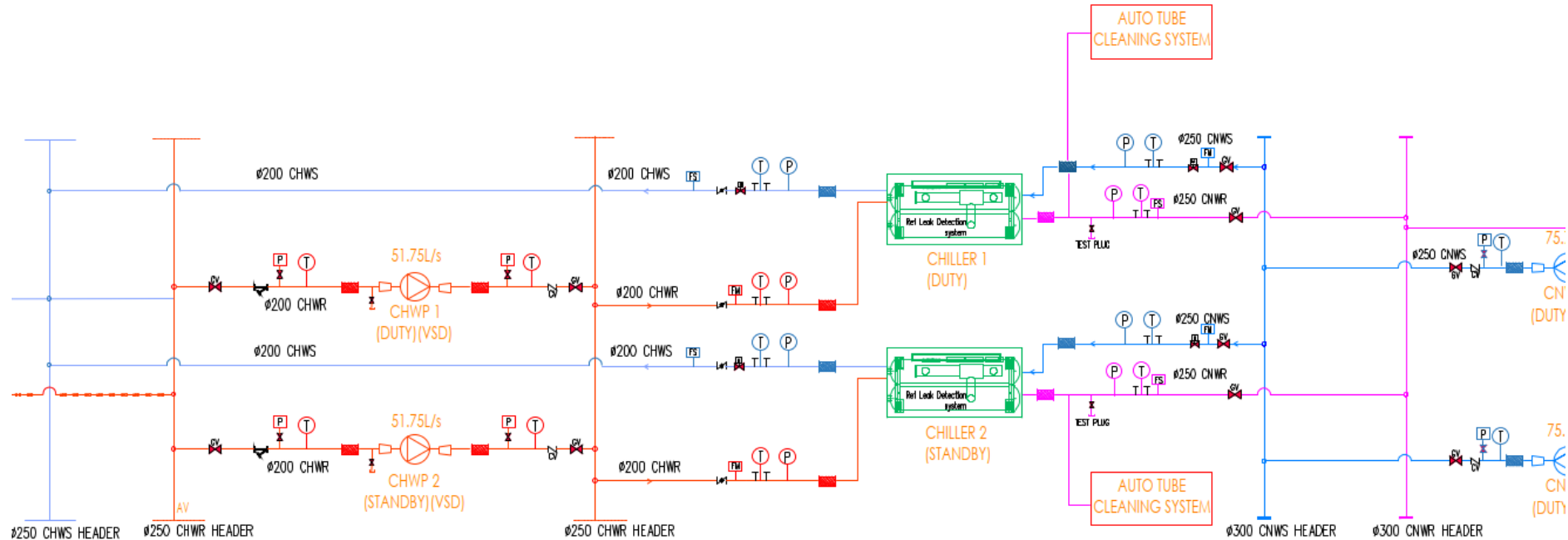
Typical Pipe Connection to Chiller

There are 4 pipe connected with Chiller

1. Chilled Water Supply Pipe (CHWS)
2. Chilled Water Return Pipe (CHWR)
3. Condenser Water Supply Pipe (CWS)
4. Condenser Water Supply Pipe (CWR)



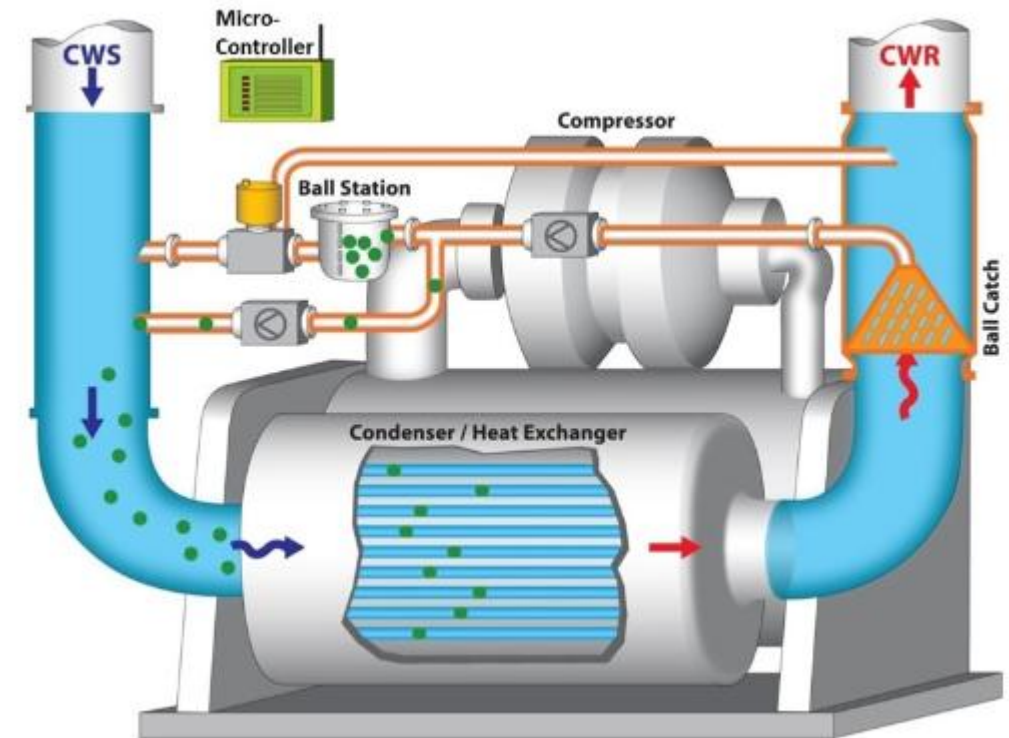
Typical Pipe Connection to Chiller



- ❖ Header Pipe : The header pipe acts as a main supply or return line that connects the chillers to the distribution system.
- ❖ Auto Tube Cleaning System is connected with condenser side of Chiller

2. Auto Tube Cleaning System

- ▶ An auto tube cleaning system for a chiller is a mechanical system that helps keep the chiller's heat transfer tubes clean and free from fouling or scaling.
- ▶ It is designed to improve the **chiller's efficiency, reduce energy consumption, and minimize maintenance requirements.**
- ▶ The auto tube cleaning system works by periodically injecting the cleaning brushes or balls into the chiller's tubes. As the chiller operates, the brushes or balls travel through the tubes, mechanically scrubbing the inner surfaces and removing any accumulated fouling or deposits.



3. Cooling Tower

Usually following are the pipe connected with Cooling Tower;

- ❖ **Condenser Water Supply Pipe (CWS)** : Supply water from Cooling Tower to Chiller
- ❖ **Condenser Water Return Pipe (CWR)** : Supply water from Chiller to Cooling Tower
- ❖ **Makeup Water Pipe**

Makeup water pipe supplies fresh water to the cooling tower system to compensate for water losses due to evaporation, drift, and bleed-off.

- ❖ **Balancing Pipe**

The primary purpose of balancing pipes is to ensure equal distribution of water flow and pressure across the cooling tower cells.

By connecting the cells together, balancing pipes allow water to be transferred between cells to balance out any differences in flow rate or pressure.

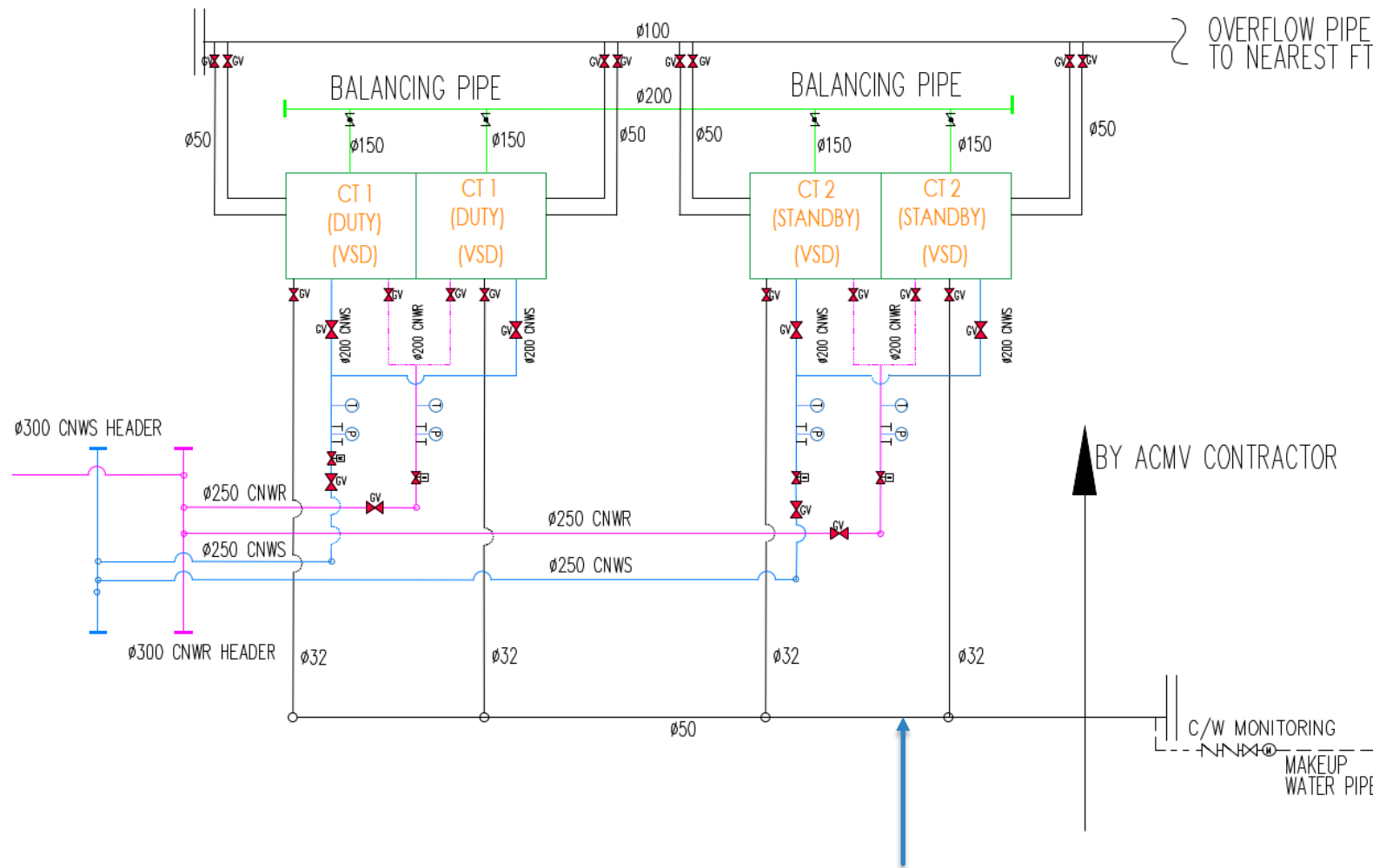
- ❖ **Overflow Pipe**

The overflow pipe in a cooling tower serves as a safety mechanism to prevent overfilling and excessive water levels within the tower.

The purpose of overflow pipe is designed to redirect excess water from the cooling tower to a safe location, such as a drain or an external water collection system. It helps prevent water from overflowing and causing damage to the cooling tower or the surrounding area.



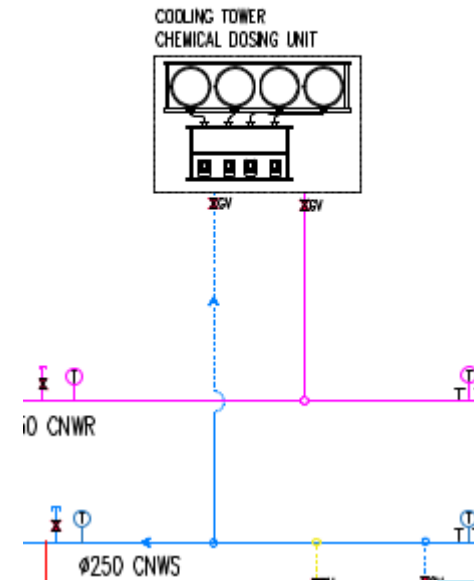
Typical Pipe Connection to Cooling Tower



Make Up Water Pipe

4. Chemical Water Treatment for Cooling Tower

- ▶ Chemical water treatment for cooling towers is a process that involves the use of chemicals to treat the water in a cooling tower system.
- ▶ Its is connected with CWS & CWR pipe as shown in the picture
- ▶ The primary goals of chemical water treatment are to **prevent scale formation, control corrosion, inhibit microbial growth, and maintain water quality.**
- ▶ By implementing proper chemical treatment, the cooling tower system can operate efficiently, minimize maintenance issues, and ensure the safety and comfort of building occupants.



SECTION A
SCALE 1:50

MECHANICAL LAYOUT DRAWING OF A CHILLED WATER PLANT ROOM. THE DRAWING SHOWS THE ARRANGEMENT OF PUMPS, PIPES, AND STRUCTURAL ELEMENTS. KEY COMPONENTS AND DIMENSIONS INCLUDE:

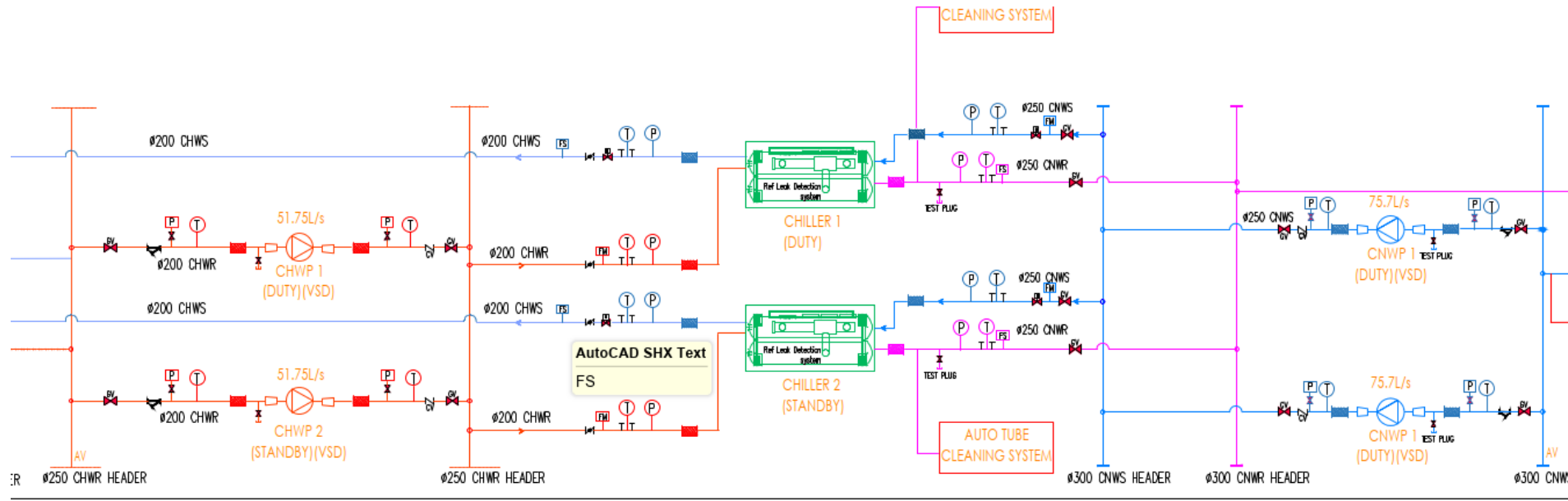
- PUMPS:** Three pumps are shown, labeled CWP-TE1 CLP-201, CWP-TE1 CLP-202, and CWP-TE1 CLP-203. Each pump is connected to a vertical riser pipe.
- PIPES:**
 - CHILLED WATER (CHW) PIPES:** Various sizes are indicated, including #250 CHWR, #400 CHWR (HEADER), #200 CHWR, and #100 CHWR.
 - CHILLED WATER RETURN (CHWR) PIPES:** Indicated by blue lines, including #250 CHWR and #400 CHWR/R PIPES CAPPED OFF FOR FUTURE CONNECTION.
 - CHILLED WATER BYPASS PIPES:** Indicated by blue lines, including #200 CHW PIPES BYPASS.
- STRUCTURAL ELEMENTS:**
 - SUPPORT CHANNEL (P) SEE TABLE (8)** and **SUPPORT CHANNEL (N) 200x90x8(t)** are shown at the top.
 - DETAIL OF CHILLED WATER PUMP BASE-FRAME** is indicated.
 - SUPPORT SHS (N) SHS 150x150x6(t)** are shown on the left.
- DIMENSIONS:**
 - Horizontal Dimensions:** 2300, 2200, 2700, 2700, 2500, 3000, 4150.
 - Vertical Dimensions:** 1376, 685, 563, 384, 930, 950, 288, 700, 950, 365.
- LEGEND:**
 - (B) #250 CHWR FROM LOAD CENTRE 2, P-WAY & BUS DEPOT**
 - (B) #250 CHWS TO LOAD CENTRE 2, P-WAY & BUS DEPOT**
 - (A) #250 CHWR FROM STABLING & RWS BUILDING**
 - (A) #250 CHWS TO STABLING & RWS BUILDING**
- OTHER NOTATIONS:**
 - TP1** (Temperature Point 1) is marked on the riser pipes.
 - SEE DETAIL 8B** and **SEE DETAIL 8A** are indicated.
 - #400 CHWS (HEADER) BOP 4150** is shown at the top.
 - #250 CHWS BOP 4975** is shown in the middle.

Technical drawing of Section B of a wellhead assembly, showing dimensions and components. The drawing includes a side view of the wellhead structure and a cross-section view of the wellbore. Key components and dimensions are labeled:

- Wellhead Components:**
 - Ø500 CWP HEADER BOP 3800
 - Ø500 CWP BOP 1850
 - Ø500 CWP HEADER BOP 645
 - Ø300 CWP BOP 740
 - BUTTERFLY VALVE
 - CHECK VALVE
 - STRAINER
 - Ø25 DRAIN PIPE
- Dimensions:**
 - Overall height: 1850
 - Wellhead height: 3800
 - Wellbore depth: 3500
 - Wellbore diameter: Ø300
 - Wellhead diameter: Ø500
 - Wellhead offset: 645
 - Wellhead offset: 1200
 - Wellhead offset: 1230
 - Wellhead offset: 350
 - Wellhead offset: 125
 - Wellhead offset: 3850
 - Wellhead offset: 600
 - Wellhead offset: 850
- Wellbore Components:**
 - CT600/W/EP (TOP TIER) BOT+3700
 - CT600/W/NP (BOTTOM TIER) BOT+3500
 - CT100/W/EP
 - CT100/W/EP BOT+75
- Other Labels:**
 - CWP-TE1 CLP-20x
 - E-STOP LOCK

SECTION B
SCALE 1:50

Typical Pipe Connection to Pump

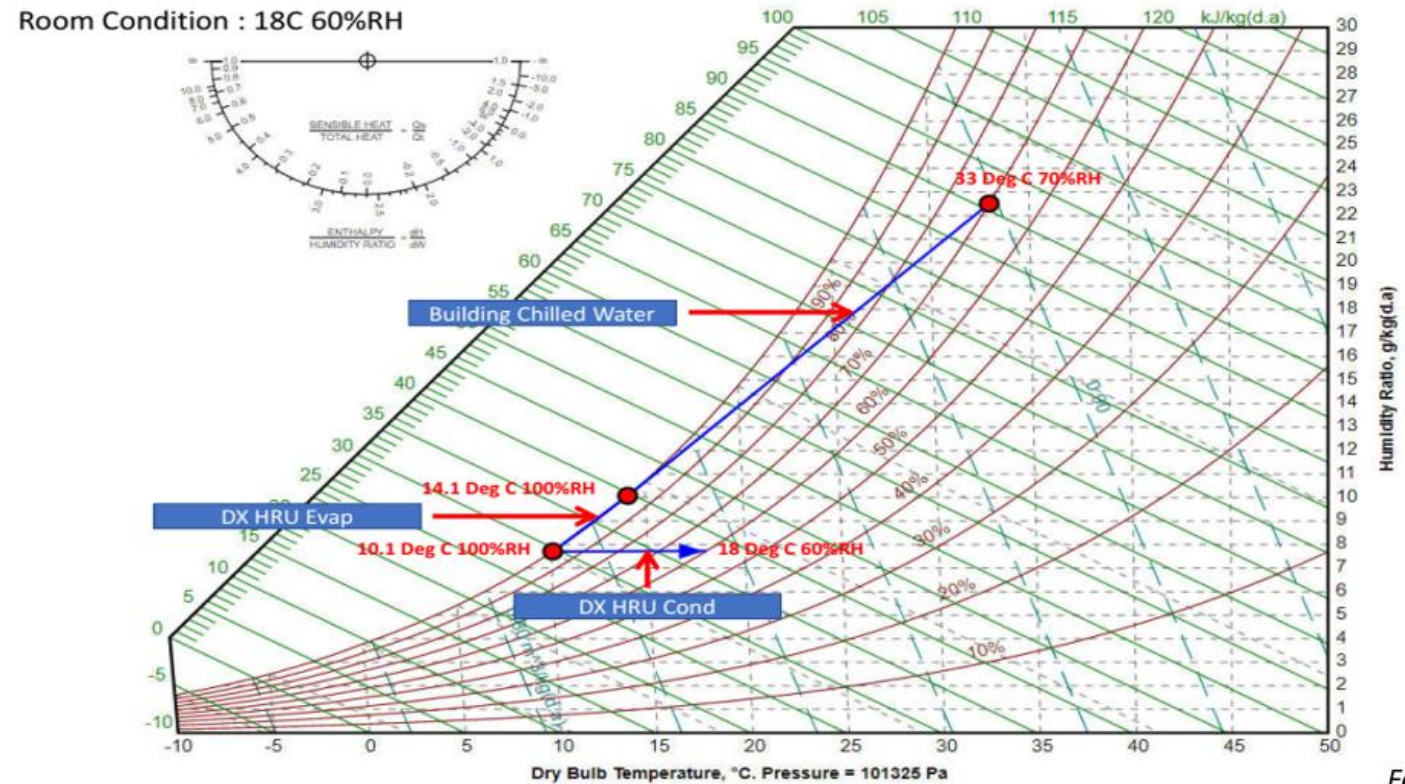


Heat Recovery Unit (HRU)

Heat recovery unit (HRU) is mostly used to achieve the room condition RH %.

There are two type of HRU

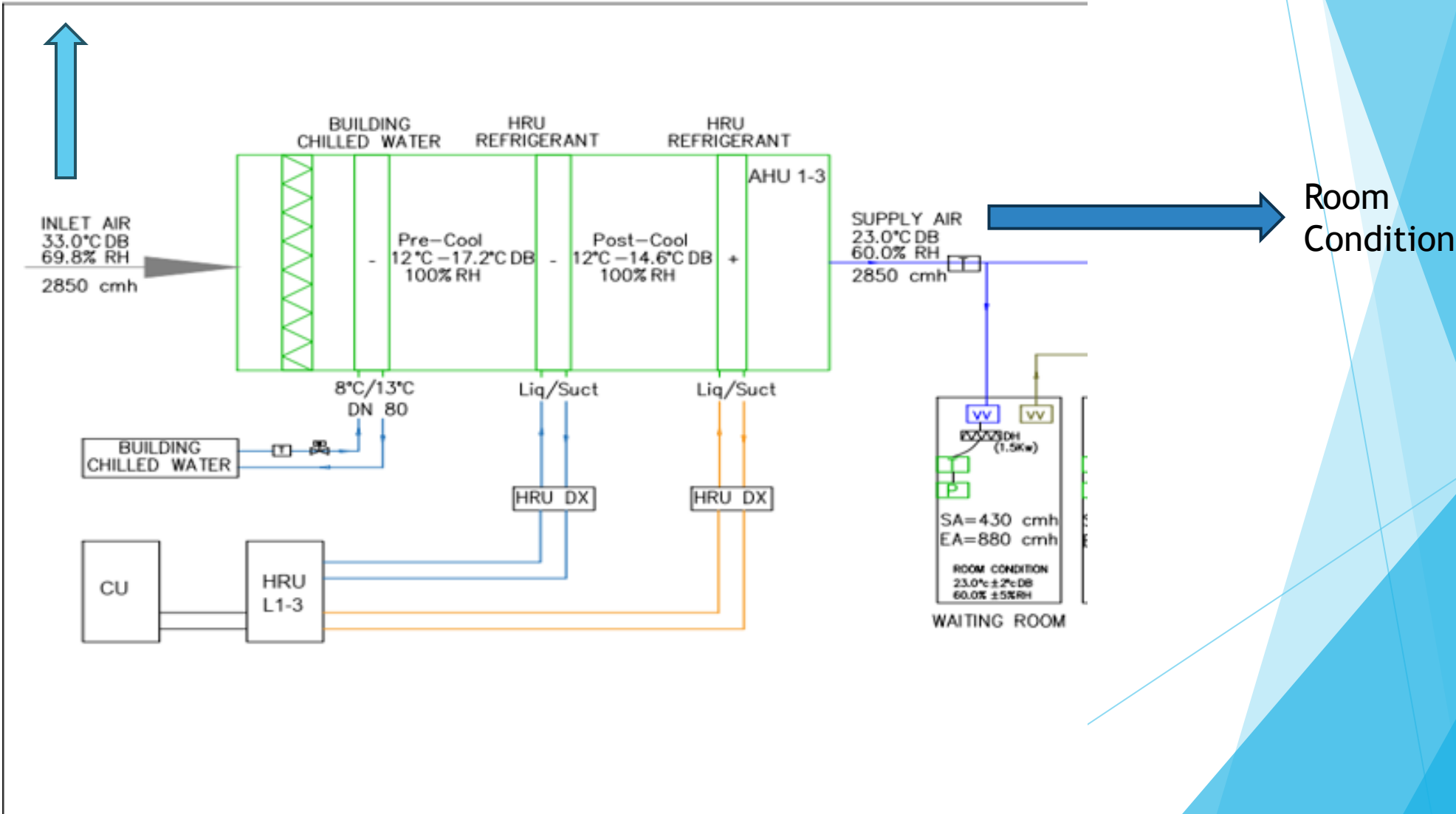
1. Chilled water HRU
2. Refrigrant HRU



For Illustration Purpose

AHU & HRU pipe connection

Out door air



Thank you !